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Key Points:

- Precipitation moisture source varies substantially with season and location in Greenland
- Large interannual variability exists for moisture source, which is partially explained by indices like the GBI
- A significant long-term trend toward more local moisture sources exists for Northwest Greenland

Supporting Information:

- Supporting Information S1
- Figure S1
- Figure S2
- Figure S3
- Figure S4
- Figure S5

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Spatial Shift of Greenland Moisture Sources Related to Enhanced Arctic Warming

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Abstract Understanding the atmospheric moisture budget of Greenland is critical for predicting future Greenland climate. However, past attempts to quantify the provenance of Greenland precipitation have been limited to certain seasons. Here we present an analysis of Greenland moisture sources using water tracers in the Goddard Institute for Space Studies climate model, which can provide source estimates for all time periods. The North Atlantic is found to be the dominant moisture source, except during summer when continental sources increase substantially. The variability is also found to be partially correlated with the Greenland Blocking Index. The key finding, however, is a long-term trend in the moisture source location for Northwest Greenland, with an increase in more locally sourced moisture over time during nonsummer months. This is at least partially related to sea ice loss in the Baffin Bay region, along with a larger increase in sea surface temperatures for the region relative to other locales.

Plain Language Summary It is still not fully clear where the precipitation that falls over Greenland actually evaporates from, and how these evaporation locations change with the climate, with previous studies being limited in terms of season or transport time. Here we use a climate model that can trace water through the atmosphere from evaporation to rainout to determine the sources of moisture for Greenland precipitation. It is found that the North Atlantic is the largest source for every season except summer, when land-based moisture sources become larger. These sources vary over time, particularly in response to the Greenland Blocking climate index. Finally, a long-term change in the moisture source for Northwest Greenland was found, with more water evaporating from locations closer to Greenland itself. This change is found to be at least partially related to the loss of sea ice and a warming in sea surface temperatures in the oceans and seas closest to Greenland, resulting in more evaporation and thus an increase in local moisture sources.

1. Introduction

The Greenland Ice Sheet (GrIS) is an important modulator of global sea level change (Church et al., 2013; Mortin et al., 2016; Shepherd et al., 2012). It is also an indicator of changes in Arctic climate, and isotope records from ice cores can be used to reconstruct past climate variability (e.g., Dansgaard, 1964; Grootes et al., 1993; Rasmussen et al., 2013). The GrIS has been losing mass since the 1980s at an accelerating rate, with the largest mass losses occurring between 2010 and 2018, adding 13 mm to global sea level over the period 1972–2018 (Mouginot et al., 2019). The surface mass balance (SMB), the balance between accumulation, runoff, net evaporation, and net sublimation, is a key driver of GrIS mass change, estimated to have contributed to 60% of 1991–2015 mass loss (van den Broeke et al., 2016).

Changes in GrIS mass balance coincide with recent increases in global temperature, with larger increases in the Arctic (Hartmann et al., 2013). The largest contribution to the recent changes in SMB has been suggested to be changes in atmospheric circulation over the ice sheet, contributing to up to 70% of the surface mass loss (Fettweis et al., 2013; Hofer et al., 2017; Mattingly et al., 2018; Tedesco et al., 2016). The changes in atmospheric circulation are characterized by anomalous anticyclonic circulation, higher 500 hPa geopotential height (Hanna et al., 2016), and a northward shift in the jet stream location (Tedesco et al., 2016). Various mechanisms for the impact of the circulation change on mass loss have been suggested, including changes in air temperature (Fettweis et al., 2013), reductions in cloud cover (Hofer et al., 2017), and atmospheric river events bringing moisture over the ice sheet (Mattingly et al., 2018).

Multiple recent studies have indicated that atmospheric moisture transport over the GrIS and other parts of the Arctic can enhance surface melting. In particular, low-level, optically thin clouds containing liquid water (Bennartz et al., 2013; Van Tricht et al., 2016) and advection of warm, moist air (Bonne et al., 2015; Mattingly et al., 2018; Neff et al., 2014) have been shown to enhance surface melt over Greenland. An extreme melt event that occurred over the GrIS in summer 2012, during which most of the ice sheet surface was covered by melt (Nghiem et al., 2012; Tedesco et al., 2013), was caused by such atmospheric conditions (e.g., Bennartz et al., 2013; Bonne et al., 2015; Mattingly et al., 2018). Thus, moisture transport in summer can influence GrIS mass loss, whereas in winter, it can impact GrIS mass gain. However, although there is usually net mass loss for Greenland in summer, snowfall still accumulates over much of the ice sheet outside of the marginal ablation zone.

Given the recent relationships between atmospheric circulation change, atmospheric moisture, and GrIS SMB, it is important to understand how the sources of moisture for the GrIS may have changed over recent years. Such changes are also important from the perspective of studying how past changes in atmospheric circulation relate to present-day and future GrIS climate variability (Sodemann & Stohl, 2013; Steen-Larsen et al., 2015). Multiple studies have shown that isotopic records from ice cores are sensitive to changes in moisture source locations and pathways and therefore changes in atmospheric circulation patterns as well (e.g., Charles et al., 1994; Lewis et al., 2013; Sodemann et al., 2008a; 2008b; Werner & Heimann, 2002). Connecting recent GrIS moisture source changes to historical changes derived from ice core reconstructions can help improve our understanding of past, present, and future variability in Greenland's climate and its role in sea level rise.

To obtain a better understanding of recent and potential future changes in GrIS moisture sources, we use a general circulation model to examine the distribution and variability of Greenland moisture sources over the 1980–2015 period. Various studies have examined variability in Greenland moisture sources using reanalysis data (Sodemann et al., 2008a; 2008b), climate model simulations (Charles et al., 1994; Langen & Vinther, 2009; Lewis et al., 2013; Werner & Heimann, 2002), and water isotope observations and simulations (Bonne et al., 2014; Steen-Larsen et al., 2011, 2013, 2014, 2017; Vinther et al., 2003), finding that Greenland moisture is mostly sourced from local oceanic regions and that the sources are sensitive to regional climatic variability, particularly the North Atlantic Oscillation (NAO). However, several key questions have not been explored in depth and are explored here. These include (1) how Greenland moisture sources vary on a seasonal basis, particularly during summer, when nearly all GrIS mass loss occurs, (2) whether there has been a trend in Greenland moisture source location over the period of accelerating GrIS mass loss, and (3) what major factors control the variability and trends in the moisture source.

2. Methods

2.1. Model Description

Here we use the Goddard Institute for Space Studies (GISS) ModelE2.1, which is similar to ModelE2 (Schmidt et al., 2014) but with a few additional improvements to clouds and moist convection (Field et al., 2014; Kim et al., 2012; Yao & Cheng, 2012). The 40-layer atmosphere has a top at 0.1 mb and a resolution of 2° latitude by 2.5° longitude. This is a lower resolution than has been used in previous studies (e.g., Singh et al., 2017), but given that the source estimates are for relatively large regions (supporting information Figure S1), the resolution should be adequate. Here we present two sets of experiments in which we run the model. Both utilize Hadley Sea Surface Temperatures and Sea Ice extent (Deser et al., 2010; Rayner et al., 2003). In the free-running atmosphere simulations, the native GISS-E2.1 winds and other atmospheric quantities are allowed to run freely. In the nudged simulation, atmospheric winds are forced to adhere to those of the MERRA2 reanalysis (Bosilovich et al., 2015; Gelaro et al., 2017). Both sets of experiments are started in 1980 and run forward until 2015. Although only a single nudged run was performed, three free-running simulations were generated with different starting months, and the ensemble average of those simulations is presented here. Finally, the moisture sources in the model are determined using the “VSD” water tracer method (Koster et al., 1986; Kelley, 2003; Lewis et al., 2010, 2013, 2014; Nusbaumer & Noone, 2018), which is described in Text S1.

2.2. Analytic Methods

We have divided Greenland into four (cardinal direction) quadrants (Figure S1), with 72° N separating North/South and −42.5° E separating East/West. To help quantify the variability in moisture source, the great-circle distance between the area-weighted geographic center of each quadrant (the black circles in

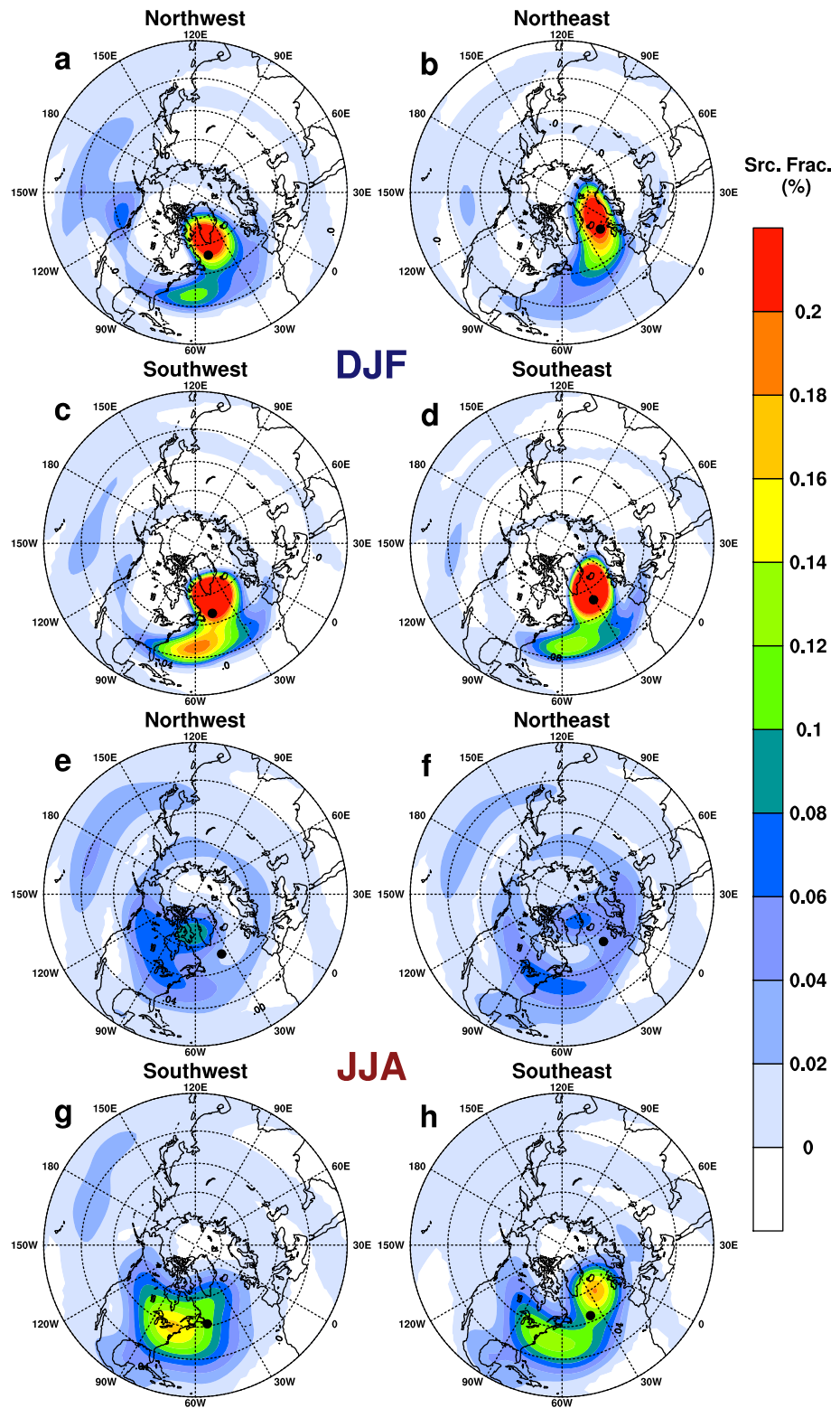


Figure 1. Climatological moisture source fraction in percent (%) averaged over December-January-February (DJF) for Northwest (a), Northeast (b), Southwest (c), and Southeast (d) Greenland, in the MERRA2-nudged simulation. The June-July-August (JJA) average moisture source is also shown for Northwest (e), Northeast (f), Southwest (g), and Southeast (h) Greenland. The moisture fraction is defined here as the fraction of total precipitation for a given Greenland quadrant that is sourced from each grid box. The black dot in each plot represents the source-weighted latitude/longitude center of each respective source distribution.

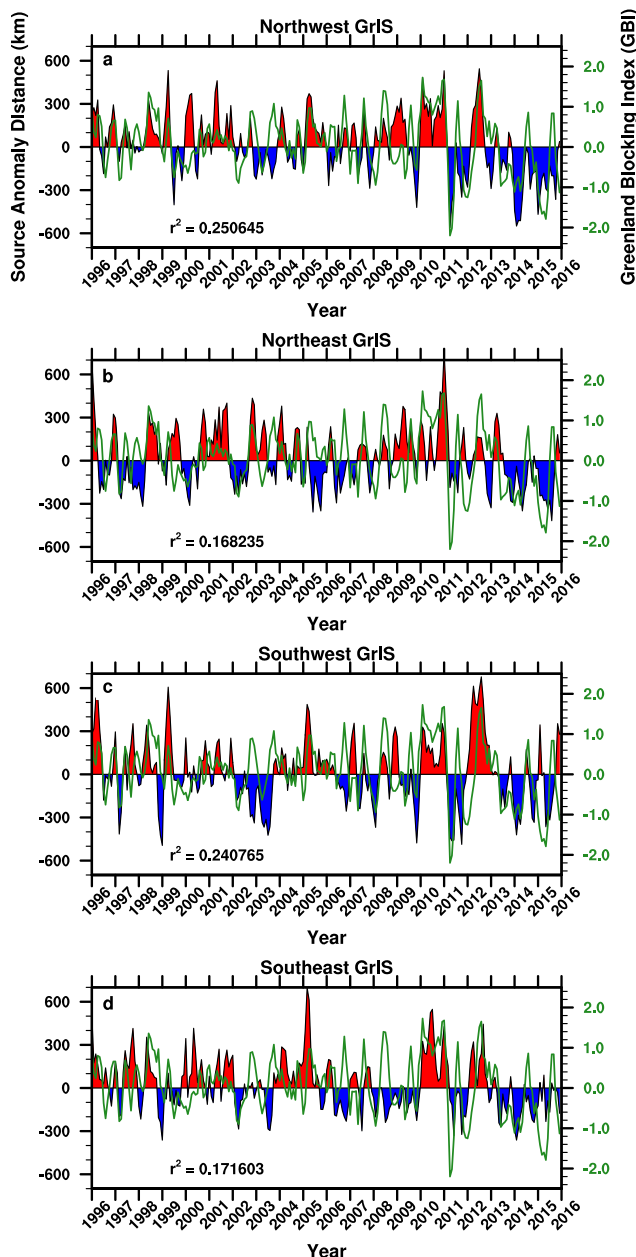


Figure 2. Detrended time series of monthly anomalies in distance (in km) between moisture source center and Greenland quadrant center for the Northwest (a), Northeast (b), Southwest (c), and Southeast (d) Greenland quadrants, over the years 1995 to 2015, for the MERRA2-nudged simulation. Red colors indicate distances that are longer than the monthly average, while blue colors indicate distances that are shorter. A time series of the Greenland Blocking Index for each month is also shown in green. A 3-month smoothing has been applied to each time series to emphasize lower-frequency variability, and the r-squared value from a simple linear regression between the smoothed GBI and the smoothed source distance anomalies for each quadrant is shown as well.

precipitation moisture source exist both seasonally and geographically for Greenland, which has implications not just for modern Greenland hydrology, but for water isotope-derived climate reconstructions as well.

A comparison between the free-running ensemble average and nudged GISS simulation reveals small differences in moisture source (Figure S3). In particular, in the free-running GISS-E2.1 ensemble, moisture in northern Greenland tends to come from more remote as opposed to local sources (Figure S3). The opposite

Figure S1) and the VSD-weighted moisture source center (e.g., the black circles in Figures 1 and S2) is calculated for each month in the simulations. A monthly average distance is then generated, and the anomalous difference between the long-term monthly average and each month's distance is determined.

Finally, several climate indices, in particular the El Niño Southern Oscillation (ENSO) as represented by the NINO3.4 SST index (Trenberth, 1997), the NAO (Barnston & Livezey, 1987), the Arctic Oscillation (Thompson & Wallace, 1998), and the Greenland Blocking Index (GBI) (Hanna et al., 2016), were examined to see if they correlated with, and thus could possibly explain, any temporal moisture source variability found in the simulations. The climate indices were determined from observations in the nudged simulation, although the climate modes were also calculated directly from model outputs, with similar results (not shown). We partition our analysis into seasonal changes, specifically boreal winter (December-January-February [DJF]), spring (March-April-May [MAM]), summer (June-July-August [JJA]), and autumn (September-October-November [SON]), with a particular focus on the autumn season, which has experienced the greatest changes.

3. Results

The climatological/long-term-average moisture sources for the four Greenland quadrants (Figure 1 and S2) vary seasonally, according to the nudged GISS simulation. The source fraction shown in the figures is defined as the fraction of total precipitation over the entire area of a given Greenland quadrant that is sourced from each grid box. The North Atlantic is the dominant source of moisture that precipitates over Greenland during boreal winter (Figure 1), spring, and autumn (Figure S2), contributing 66.7% of moisture in December-January-February and greater than 50% during spring and autumn, which generally matches previous results (e.g., Sodemann et al., 2008a). A substantial amount of moisture is also sourced from the North Pacific during these months (with a total fraction between 11% and 14%), which also matches previous studies (Langen & Vinther, 2009), although the value may be biased high given the model resolution and subsequent topography. In contrast, during boreal summer (Figure 1), a large fraction of moisture (45% to 53% depending on the quadrant) is sourced from continental regions.

Along with seasonal variations, the moisture source of precipitation varies noticeably with the geographic location on Greenland (Figure 1). Specifically, much of the Western Greenland quadrants' moisture is sourced from the Labrador Sea region north of Newfoundland, Canada (Figures 1a, 1c, 1e, and 1g), whereas much of the Eastern quadrants' moisture is sourced from the seas and ocean around Iceland, including the North and Norwegian seas (Figures 1b, 1d, 1f, and 1h). Also, although perhaps not surprisingly, Southern Greenland's moisture tends to originate from farther south, in particular from the subtropical North Atlantic (Figures 1c, 1d, 1g, and 1h), which matches previous studies (e.g., Bonne et al., 2014; Sodemann et al., 2008a). Thus, substantial variations in pre-

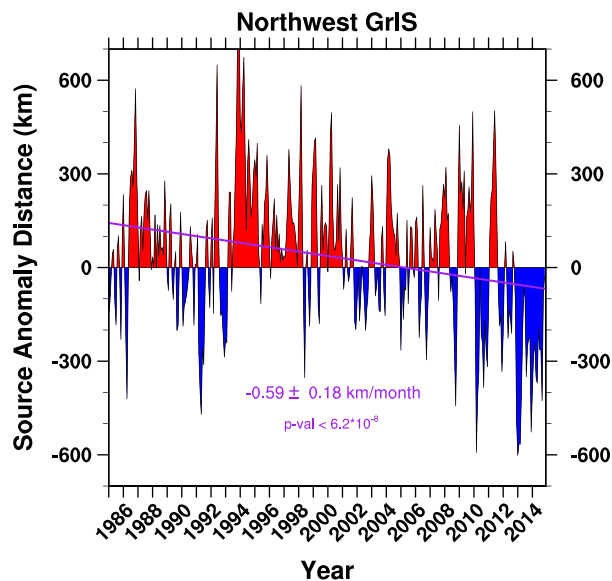


Figure 3. The time series of monthly anomalies in distance between moisture source center and the Northwest Greenland quadrant center for the Years 1985 to 2015 in the MERRA2-nudged simulation. Red colors indicate distances that are longer than the monthly average, while blue colors indicate distances that are shorter. Also plotted is the long-term trend as calculated using a simple linear regression between time and the source distance, with the slope shown in km per month. A 3-month smoothing has been applied to the distance anomaly time series to emphasize the lower-frequency variability.

of the variance and thus are likely not the primary drivers of the remaining variability, although they may have a more significant influence in specific seasons, which will be examined in future work.

Along with this interannual variability in moisture source distance, the full Northwest quadrant time series from the MERRA2-nudged simulation also contains a long-term trend, which is significant at a p -value of 6×10^{-8} according to both an F -test and a nonparametric Mann-Kendall trend test (Figure 3). This indicates that the moisture source distance is decreasing over time, or that the moisture sources for Northwest Greenland are becoming more local. The same holds for the free-running GISS ensemble simulations, which also have a calculated negative trend, although not quite as large or significant (-0.22 to -0.26 km/month, p -value ≈ 0.002 , not shown). The other quadrants appear to have trends as well, but they were not statistically significant. We also examined trends in the Northwest quadrant by season, to try to understand the cause of the trend (Figure S4). Although all seasons except JJA have a decreasing trend, the trend is largest in SON, with the smallest p -value (~ 0.02). Thus, the overall trend appears to be dominated by changes in the autumn season. Although this season is not as important for Greenland SMB changes, it still could have an impact on the water isotopes, and provides an opportunity to directly quantify the influence of moisture source changes on Greenland's isotopic record. This result also matches the predictions of Singh et al. (2017), which found increased local sources for polar regions in a warmer world.

To determine the spatial characteristics of this trend, the change in moisture source fraction between the last five SON seasons in the nudged simulation (2011–2015) and the first five seasons (1985–1989) is shown in Figure 4a, with Figure S5a showing the same results, but using 10-season averages. The figures show that there has been a shift in moisture source over this period away from the North Atlantic and into the Baffin Bay region, which explains the decrease in moisture source distance for Northwest Greenland. The free-running ensemble also exhibits a shift to more local sources (Figure S5b) but from the Pacific to the Atlantic basins, as opposed to a poleward shift. This could be due to differences in low-frequency internal variability between the nudged simulation and ensemble, but could also indicate a possible error or bias in the way atmospheric circulation changes, or surface flux changes, in this region are simulated in GISS-E2.1, which in turn will bias the moisture source results.

is true for southern Greenland, suggesting that the impact of nudging differs between the middle and high latitudes.

Along with the seasonal cycle, Greenland moisture sources have also been found to vary interannually (e.g., Sodemann et al., 2008a). To help quantify interannual variability in moisture source, we computed monthly time series of the distance of the VSD-weighted moisture source center from the geographic center of each Greenland quadrant (Figure 2). The values shown in Figure 2 have been detrended and smoothed using a 3-month running average to emphasize lower-frequency variability. It can be seen that there is a substantial amount of variability from monthly to interannual scales in the moisture sources for each quadrant. In general, the variability is coherent between the four quadrants, indicating that the drivers of this variability are large enough to influence Greenland as a whole. However, it should also be noted that the magnitude of this variability is not the same among the quadrants, indicating that some regions in Greenland may be more sensitive to internal climate variability than others, at least in terms of their moisture sources. These differences, though, are likely sensitive to the resolution of the GrIS itself and should be reexamined with higher-resolution models.

Out of all the climate indices examined, the GBI (shown in Figure 2 alongside distance anomalies for the nudged simulation) was found to exhibit the best correlation with moisture source distance anomalies, but only appears to explain 17–25% of the variance. The NAO shows a similar correlation, but it is also highly correlated with the GBI itself ($r^2 = 0.79$) and thus may not provide much additional predictability. The AO ($r^2 = 0.08$ to 0.12) and El Niño Southern Oscillation ($r^2 = 0$ to 0.06) explain very little

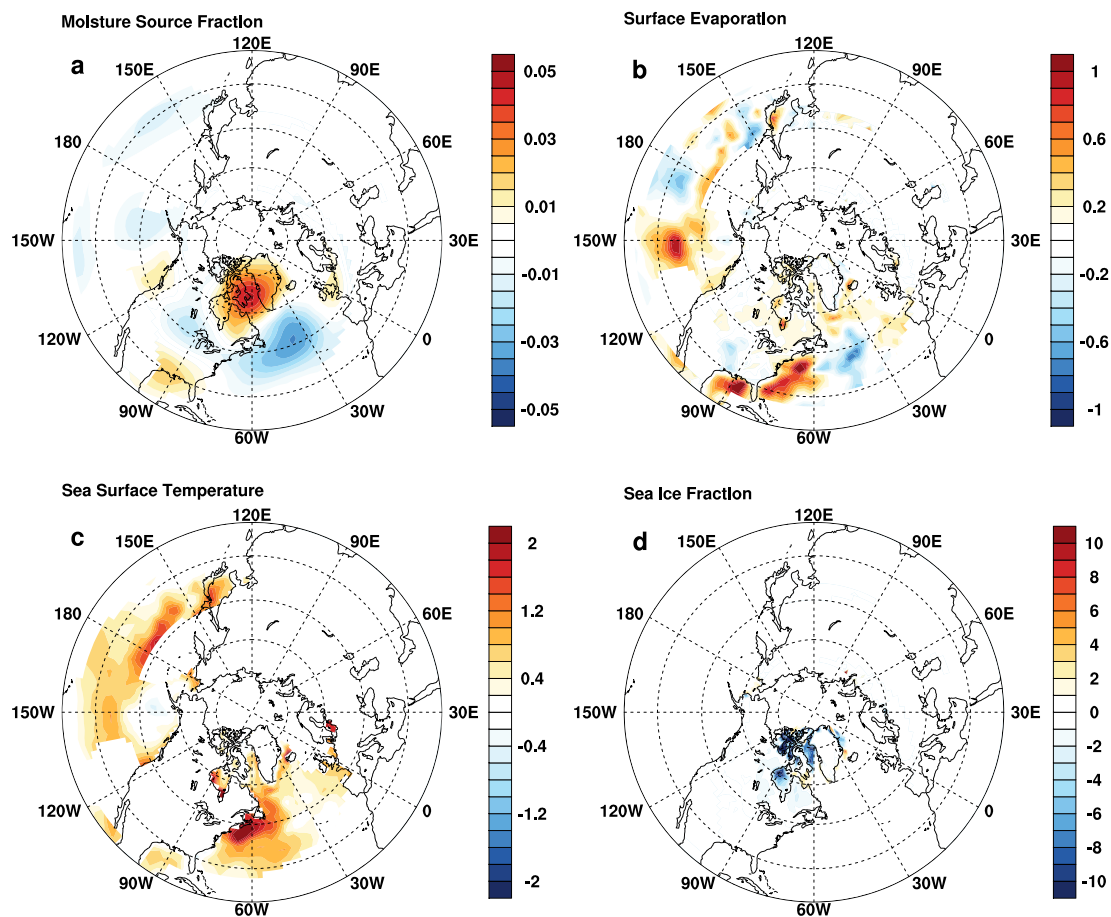


Figure 4. Difference in moisture source fraction in percent for Northwest Greenland (a), as well as surface evaporation in mm/day (b), sea surface temperature in Kelvin (c), and sea ice fraction in percent (d) between the first five September-October-November seasons (1985–1989) and the last five September-October-November seasons (2011–2015) of the MERRA2-nudged simulation. For all plots, orange colors indicate values that are increasing over time, while blue colors indicate values that are decreasing over time. Difference values are also masked to only show changes in regions with an average moisture source fraction of 0.01% or higher.

To try and determine the underlying driver of this change, we also examined the changes in surface evaporation, sea surface temperatures (SSTs), and sea ice fraction over the same time period as the moisture source fraction changes (Figures 4b–4d, respectively, along with Figures S5c and S5d). The SSTs and sea ice fractions are prescribed as boundary conditions for both the free-running ensemble and the nudged GISS simulation, and both show noticeable trends over this time period, along with the surface evaporation. In particular, sea ice loss in the Baffin Bay region, along with an increase in SSTs, results in increased surface evaporation in the region. In addition, a lack of SST warming in the eastern North Atlantic (i.e., the “warming hole”; Drijfhout et al., 2012) and Northeast Pacific generates a decrease in the evaporation in those regions relative to evaporation elsewhere, resulting in a decrease in the fraction of moisture sourced from there. This increase in Baffin Bay evaporation, combined with the relative decrease in moisture sourced from the warming-hole region and northeast Pacific, may be at least partially generating the moisture source shifts found here. However, circulation shifts may also be playing a role, including influencing the evaporation changes themselves, and will need to be examined in future work.

4. Conclusions

Recent trends in the climate of the Arctic may be impacting, or shifting, the moisture sources for Greenland. Changes in Greenland’s moisture sources could modify the heat and energy budget of Greenland itself, impacting ice loss (or gain). Changes in moisture sources can also modify the water isotope ratios of Greenland snow and thus the values recorded in Greenland ice cores, which are frequently used

for paleoclimate reconstructions. Thus it is vital to determine what the sources of moisture for Greenland are and how they might be changing with global warming.

Using a water tracer-enabled version of GISS ModelE2.1, nudged to MERRA2 horizontal winds, we calculated the average moisture source for four different Greenland quadrants over 1980–2015 during all seasons. It was found that the North Atlantic is the dominant moisture source for all of Greenland for every season except JJA, when continental moisture increases. This implies that the North Atlantic may contribute the most mass to the GrIS itself, as JJA is a time period of mass loss, although the JJA moisture sources can influence that mass loss by impacting clouds and radiation. The isotope ratios may also be impacted throughout the year due to postdepositional exchange (Steen-Larsen et al., 2014). Moisture source varies with time as well, with the interannual source variability correlating most strongly with the GBI. However, much of the interannual variability remains unexplained, and will be more closely examined in future work.

A key finding of the analysis is a long-term trend in moisture source for Northwest Greenland, with the average source region becoming more local over time. This appears to be generated by an increase in evaporation in Baffin Bay, and a relative decrease in the North Atlantic and Northeast Pacific, coinciding with long-term changes in SSTs and sea ice fraction. A similar local source increase is found in free-running simulations, although the shift is more zonal, from the Pacific to Atlantic basins. This indicates that SSTs and sea ice cover may be important factors for the changes, although circulation changes may still be modulating the actual source location shifts. The long-term trend in the moisture source could have major implications for both the climate of Greenland and for the isotope ratios of precipitation, which could in turn influence paleoclimate reconstructions using Greenland ice core data.

It should be noted that using a single model creates several methodological limitations that need to be addressed. In particular, the model has a lower horizontal resolution compared to other, regional, models (e.g., Gallée & Schayes, 1994). This means that the topography, and thus hydrology, of Greenland may not be properly simulated (Alexander et al., 2019). It also means that continental features such as mountain ranges are not properly represented either, which could result in overly large cross-continental moisture transport.

Future GISS simulations with water isotope tracers, along with simulations using alternative, higher-resolution water tracer and isotope-enabled climate models, will be performed to test the robustness of this result, and the hypothesis that the hydrologic cycle of Greenland, and the Arctic as a whole, is changing in substantial and fundamental ways.

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